Janlita Technology

Speaker Department

Customer Name:

Customer Model:

Customer P/N:

Janlita P/N: JB2831NR04W03-W06

Versions：2.0

JANLITA	CUSTOMER APPROVAL
湖南捷力泰科技有限公司 SPK 工程部专用章	

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Revision History

Date	Revision level	Description of changes
2025-06-23	1.0	Initial release
2025-07-04	2.0	外壳优化，性能改善

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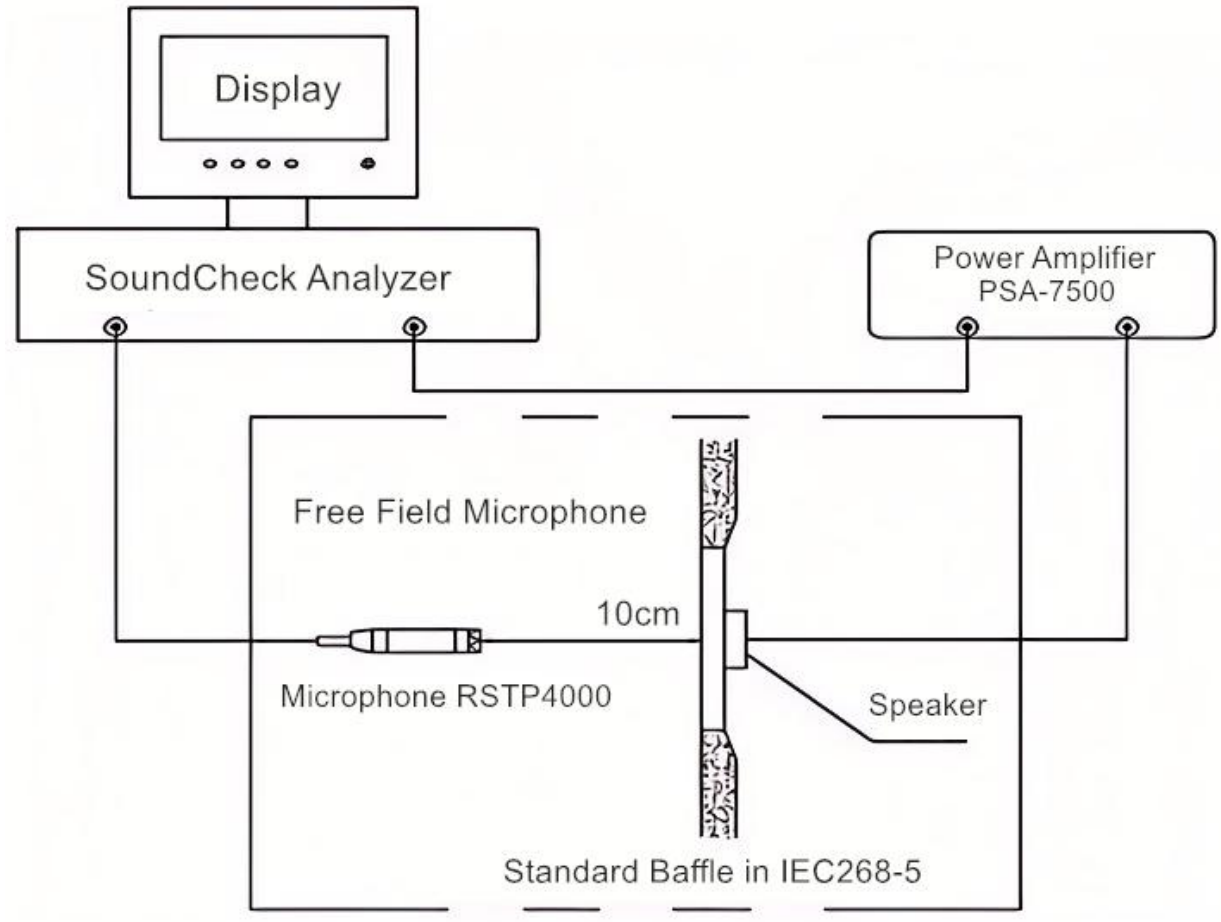
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1. Product Specification

<table>
<tr><td>1.1</td><td>Scope</td><td>This specification cover our product of speaker unit for use in GPS,phone,smart home,alarm system and calling system.</td></tr>
<tr><td>1.2</td><td>Shape size</td><td>L44*W31*H15.2mm</td></tr>
<tr><td>1.3</td><td>Net weight</td><td>19.5g±10%</td></tr>
<tr><td>1.4</td><td>Voice coil diameter</td><td>Φ16.28mm</td></tr>
<tr><td>1.5</td><td>Magnet size</td><td>Φ15.5mm</td></tr>
<tr><td>1.6</td><td>Environmental</td><td>Meet RoHS requirements</td></tr>
<tr><td>1.7</td><td>Sound pressure level(SPL)</td><td><table><tr><td>1.2kHz</td><td>1.5kHz</td><td>2kHz</td><td>2.5kHz</td></tr><tr><td colspan="4">Average：94±3 dB</td></tr></table>3W/10cm</td></tr>
<tr><td>1.8</td><td>Dc resistance</td><td>/</td></tr>
<tr><td>1.9</td><td>Impedance</td><td>4.0±15% ohm (at 1kHz/3W)</td></tr>
<tr><td>1.10</td><td>Resonant frequency(F0)</td><td>530±20% Hz
3W</td></tr>
<tr><td>1.11</td><td>Total harmonic distortion</td><td>10% Max . at 1kHz
3W/10cm</td></tr>
<tr><td>1.12</td><td>Frequency range</td><td>F0~18kHz</td></tr>
<tr><td>1.13</td><td>Rate noise power</td><td>3W</td></tr>
<tr><td>1.14</td><td>Max input power</td><td>4W</td></tr>
<tr><td>1.15</td><td>Audible noise
(Pure sound detection)</td><td>The input power shall be set at 3.46V. Using an audio oscillator, sweep from 400Hz to 3kHz. Test 3~5 times. During these sweeps, no buzz, rub, rattle or bottoming sound should perceive by the operator.</td></tr>
<tr><td>1.16</td><td>Polarity</td><td>When a positive D.C current is applied to the voice coil terminal marked (＋).the diaphragm shall move forward</td></tr>
<tr><td>1.17</td><td>Grade of waterproof</td><td>/</td></tr>
<tr><td>1.18</td><td>T/S parameter</td><td>/</td></tr>
</table>

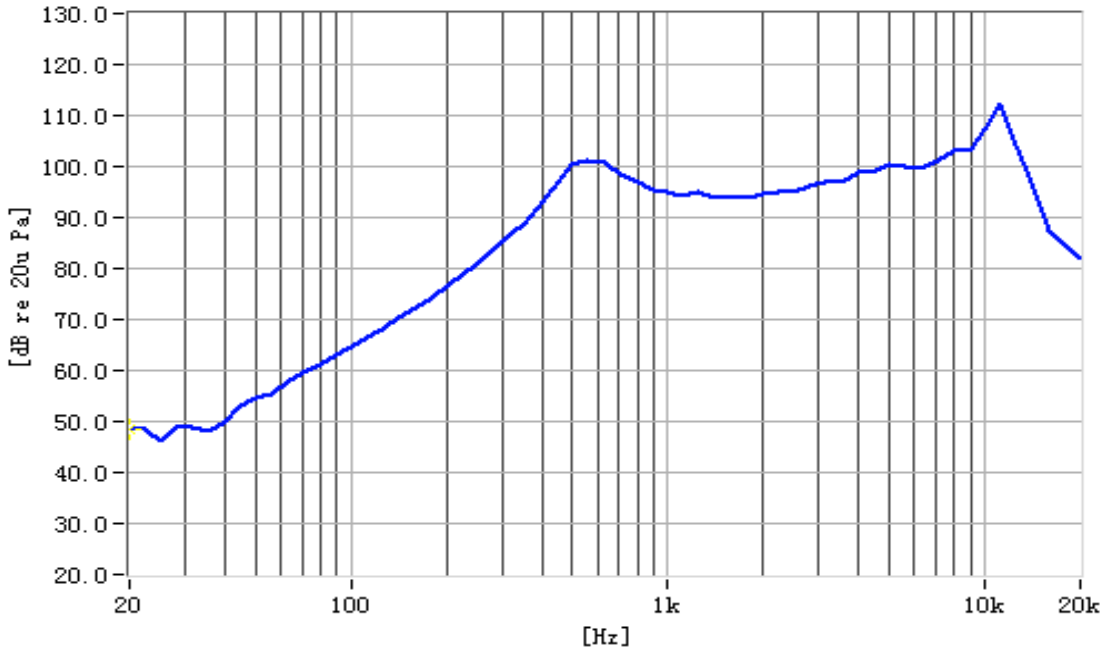
2. Measurement Methods

Test Climatic Condition	
Ambient Temperature	15~35℃, preferably at 23℃
Relative Humidity	25%~75%, preferably at 50%
Air pressure	86kPa~106kPa
General Requirements	
Operational Temperature Range	-25℃~+55℃
Storage Temperature Range	-25℃~+55℃
Devices Layout	
Chamber Type	Fully Anechoic Chamber
Microphone	Free-Field Microphone
Distance between Mic and speaker	10cm on axis of speaker
Measuring instrument	SoundCheck Test sequence R20 measurement standard (1/6 octave), sweep sine wave range from 20Hz to 20kHz.

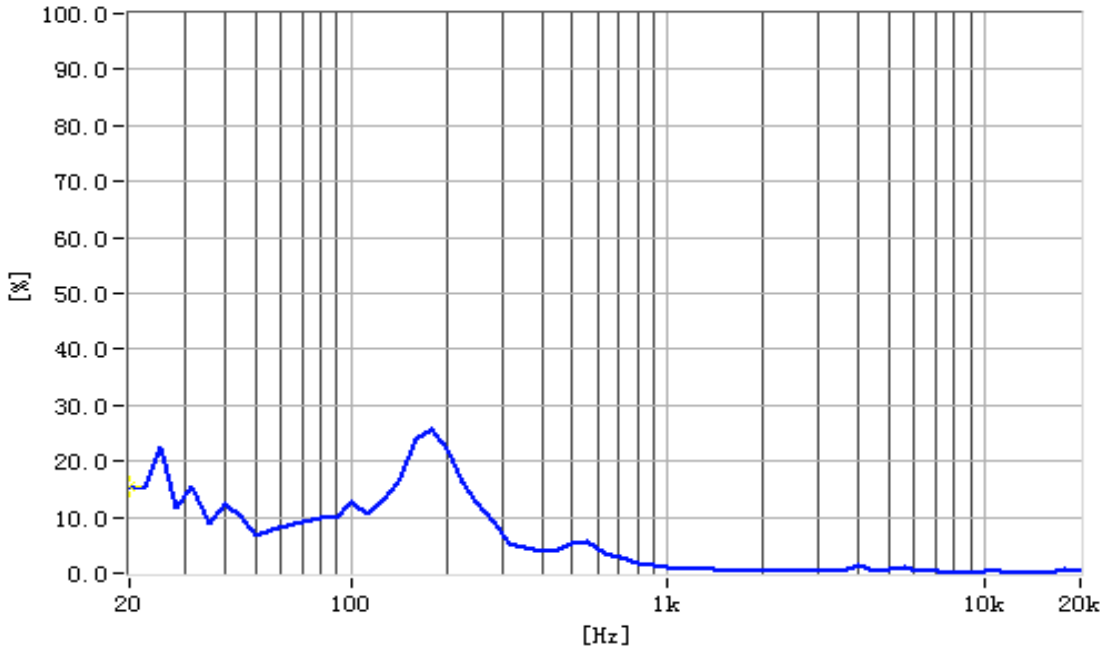


3. Acoustical Performance

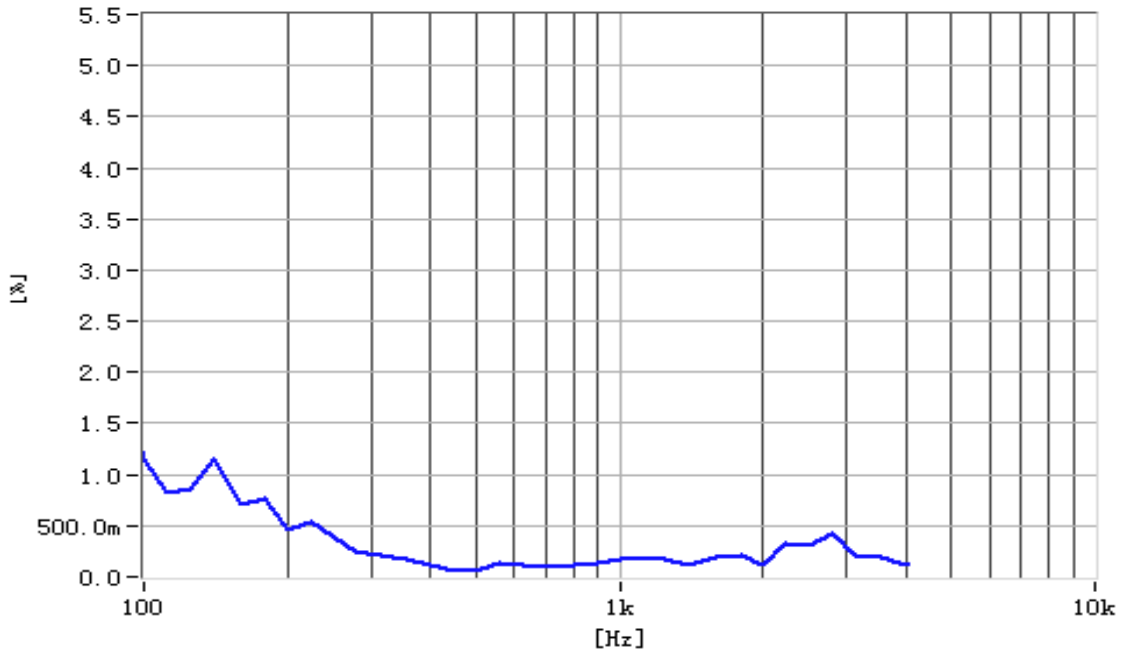
3.1 Frequency response curve



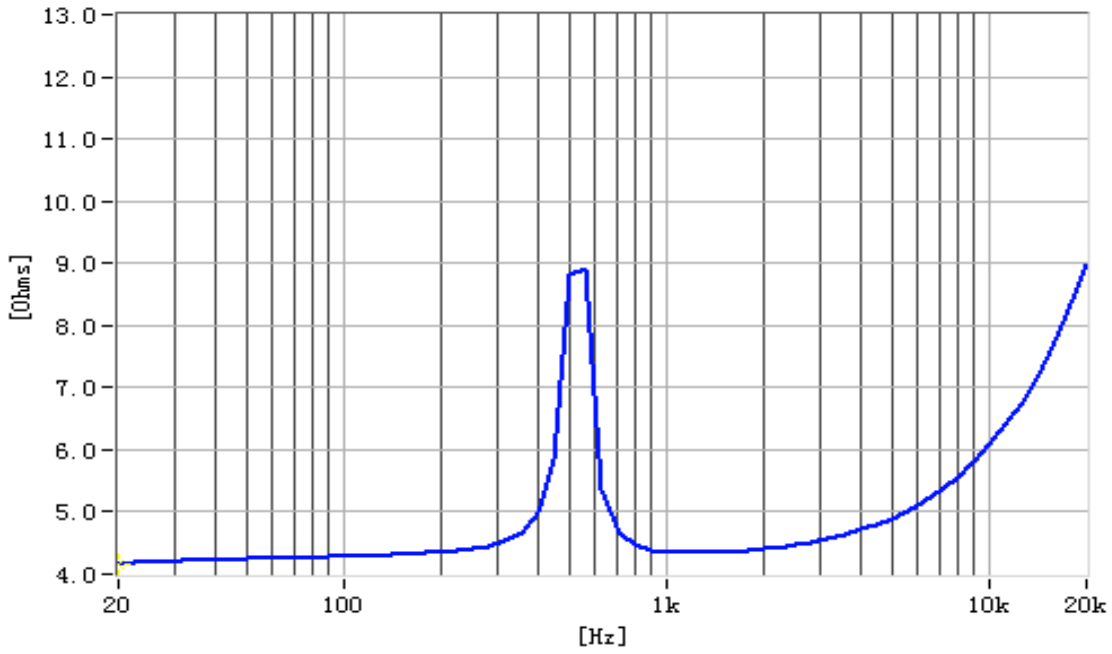
3.2 Total Harmonic Distortion curve



3.3 Rub & Buzz curve

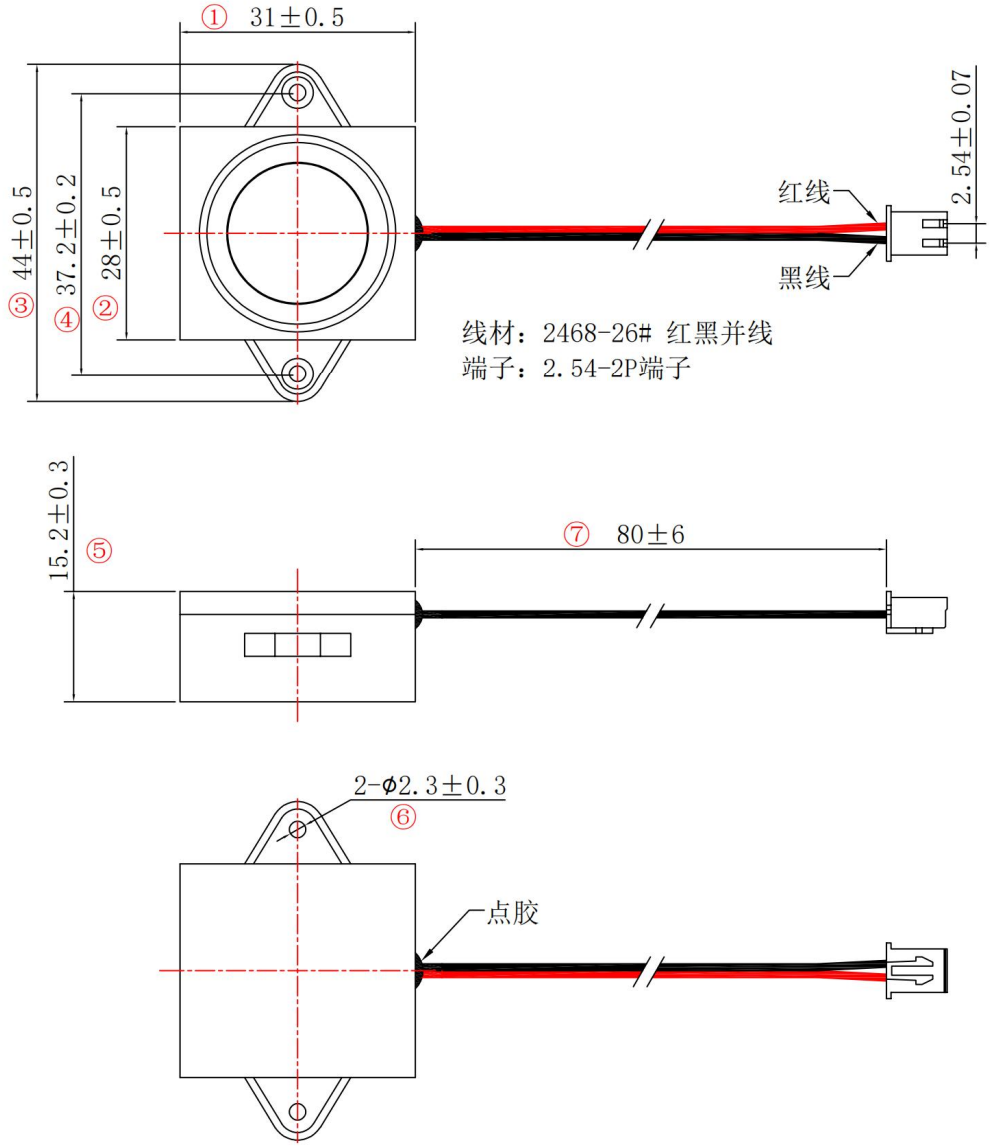


3.4 Impedance curve



4. Mechanical Layout and Dimensions

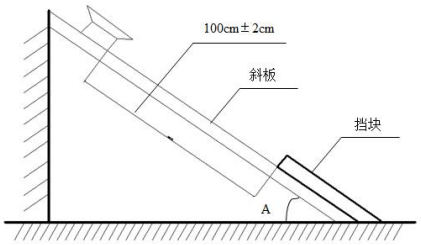
unit : mm(注：序号尺寸为关键尺寸)



No.	Item	Material	Qty.	Remark
1	支架 Bracket	ABS+PC	1	
2	U 杯 Yoke	SPCC	1	
3	磁铁 Magnet	NdFeB	1	
4	华司 Washer	SPCC	1	
5	音圈 Voice coil	Copper	1	
6	鼓纸 Diaphragm	PU	1	
7	帽子 Dust Cap	PET	1	
8	端子板 Terminal	PCB	2	
9	外壳 Casing	ABS+PC	1	

5. Reliability test

The sound pressure as specified shall neither deviate more than ±3dB from the initial value, nor any significant damage after any of following testing.

No.	Item 项目	Condition 条件
5-1	Load Test	Input power：3W (3.46V) Noise Signal：White noise from F0 to 20kHz Duration：96 hours Duration of recovery：Wait for 2 hours at room temperature
5-2	Max.Power Test	Input power：4W (4V) Noise Signal：White noise from F0 to 20kHz Duration: 1min on, 2min off, 10 cycles Duration of recovery：Wait for 2 hours at room temperature
5-3	High temperature	Temperature：60±3℃ Duration：24 hours Duration of Recovery：Wait for 2 hours at room temperature
5-4	Low temperature	Temperature：-20±3℃ Duration：24 hours Duration of recovery：Wait for 2 hours at room temperature
5-5	High humidity & high temperature	Temperature：40±3℃ Humidity：90~95 % Duration：48 hours Duration of recovery：Wait for 2 hours at room temperature
5-6	Temperature cycle	Temperature/Duration: +60±3℃ 2H→+25±3℃ 2H→ -20±3℃ 2H (This is one cycle,cycles:2 times) Duration of recovery：Wait for 2 hours at room temperature
5-7	Vibration Test	Vibrate：10~55Hz/min,Amp:1.5mm Duration : 2 hours
5-8	Drop Test	Straight line length：100cm±2cm Angle (A)：15°±2° Times：5 times 

6. Packaging

